

Thermodynamics

Lecture 2A:

- Temperature & pressure
- Energy, Energy Transfer

Prof. Claudio Hail (he/him)

After this lecture

- Discuss definitions of density, temperature and pressure.
- Understand the concept of energy and its different forms.
- Learn about the nature of internal energy.
- Concept of energy transfer by heat and work.

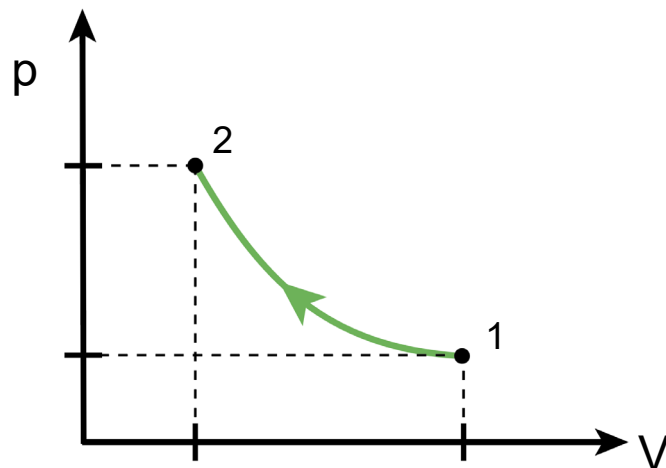
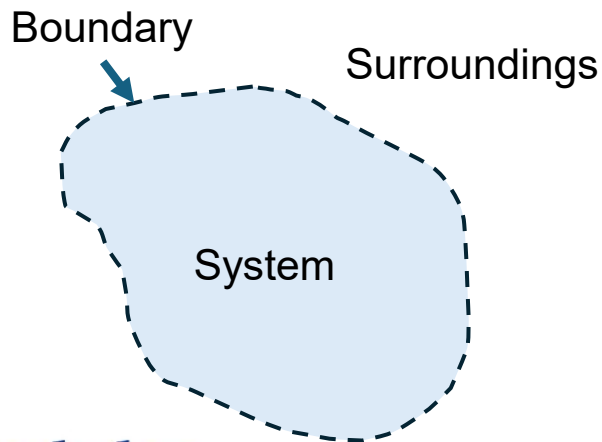
Learning objectives

What you will be able to do after this course:

- I. *Knowledge:* Write the first and second laws of thermodynamics.**
- II. Comprehension:* Describe the first and second laws in the student's own language.
- III. Application:* Solve simple single-answer problems using the first law.
- IV. Analysis:* Solve problems requiring both the first and second laws.
- V. Synthesis:* Develop schemes to improve the efficiency or performance of thermodynamic systems, e.g., compressors, turbines, heat pumps, etc.

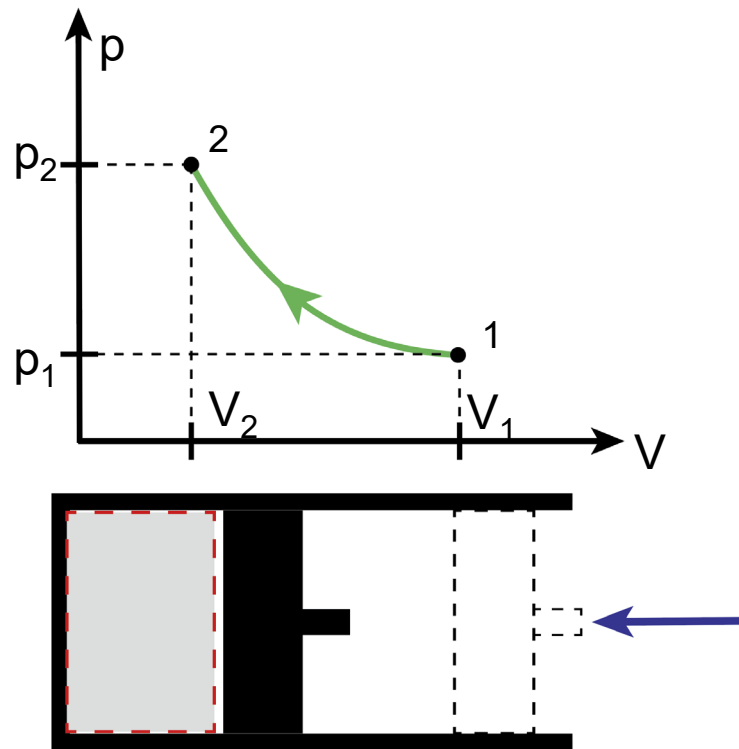
Recap Lecture 1B

- Define thermodynamic System: open, isolated, closed.
- Thermodynamic state and state properties.
- Thermodynamic processes and cycles

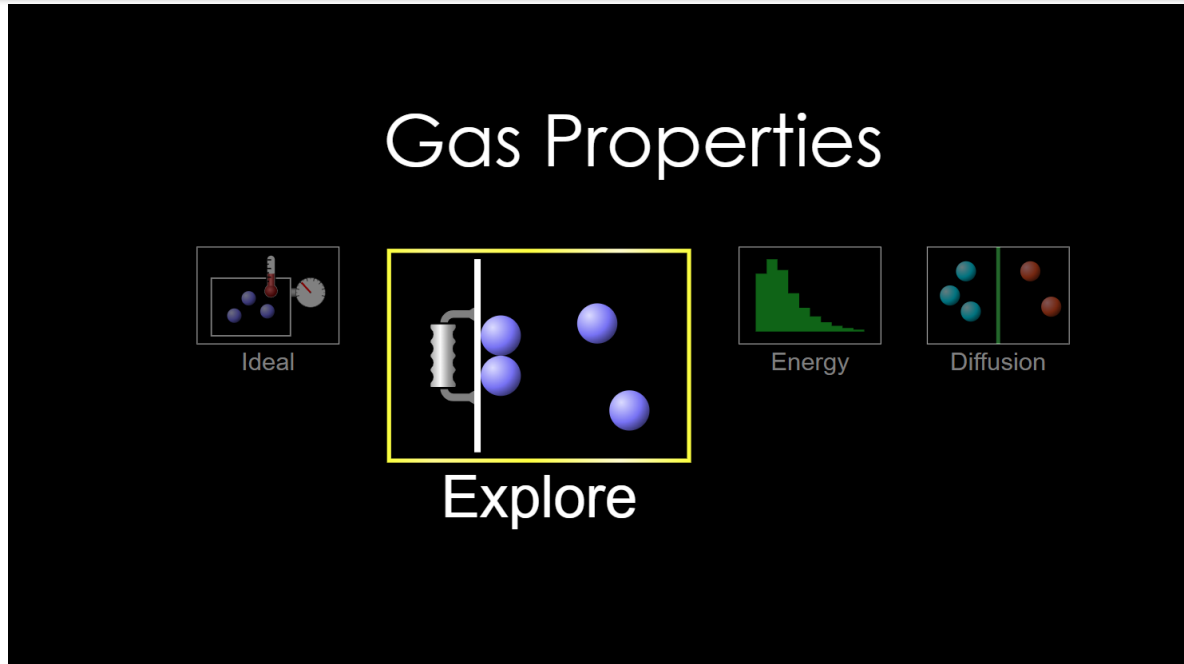


Thermodynamic process (II)

- Quasistatic process:
 - Process runs sufficiently slow so that systems passes through a series of equilibrium states.
- Non-quasistatic process:
 - Process runs too fast for equilibrium to be reached.

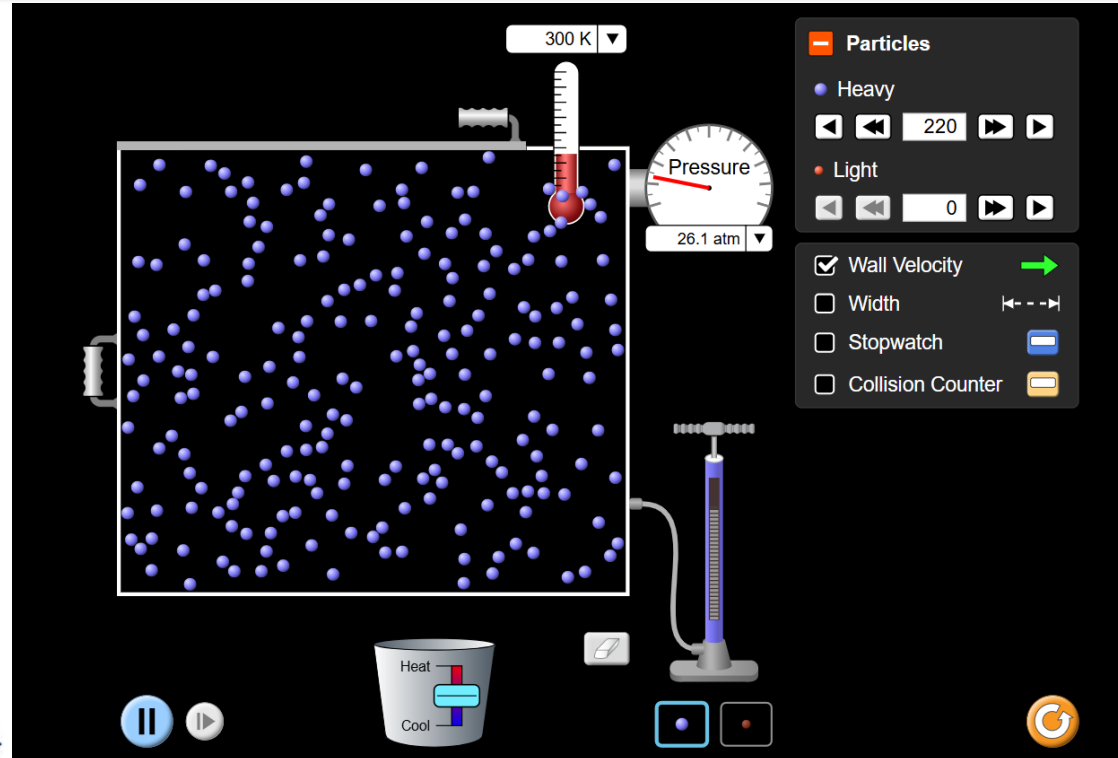


Simulation: Gas in a cylinder



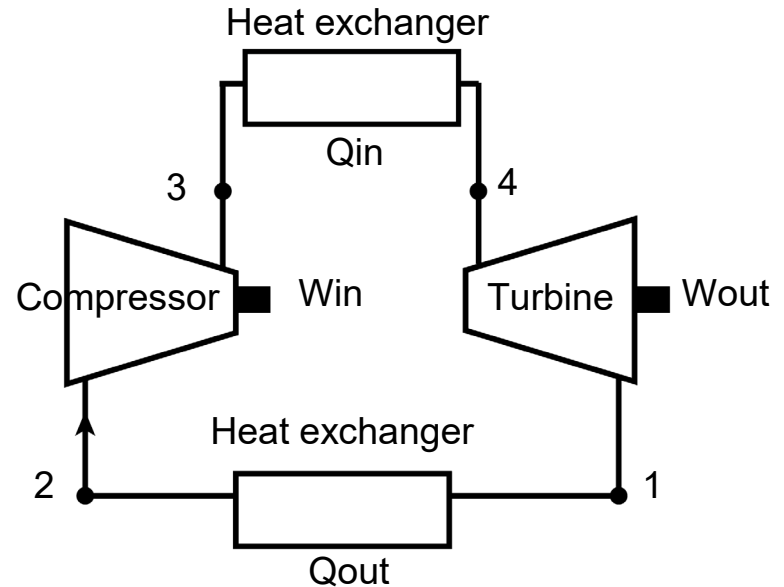
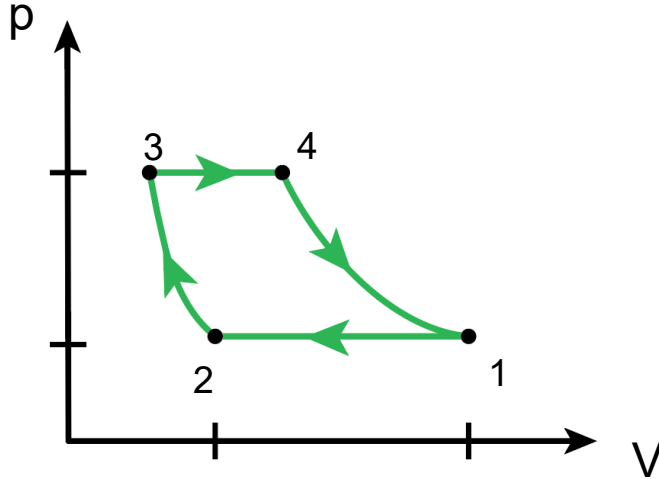
https://phet.colorado.edu/sims/html/gas-properties/latest/gas-properties_all.html

Simulation: Gas in a cylinder



Thermodynamic Cycles

- System undergoing a series of thermodynamic processes and returning to the initial state.



Density

- Density: Mass per volume

$$\rho = \frac{m}{V} = \left[\frac{kg}{m^3} \right]$$

- Specific Volume:

$$v = \frac{V}{m} = \frac{1}{\rho} = \left[\frac{m^3}{kg} \right]$$

- Generally:

$$\rho = \rho(p, T)$$

- Incompressible fluid:

$$\rho(p, T) \approx \rho_{inc}(T)$$

Temperature

- A measure of how hot or cold
- **Celsius:** ice point 0°C , steam point 100°C
- **Fahrenheit:** ice point w salt 0°F , human body 96°F
- **Kelvin:** absolute zero 0 K

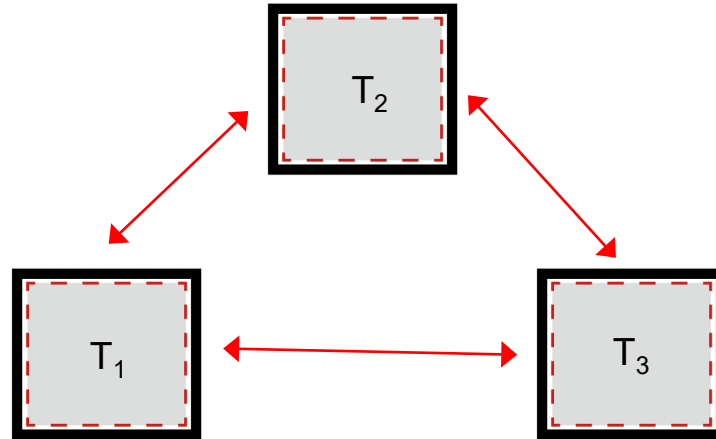
$$T(K) = T(^{\circ}\text{C}) + 273.15$$

$$T(K) = \frac{T(^{\circ}\text{F}) - 32}{1.8} + 273.15$$

$$\Delta T(K) = \Delta T(^{\circ}\text{C})$$

0th law of thermodynamics

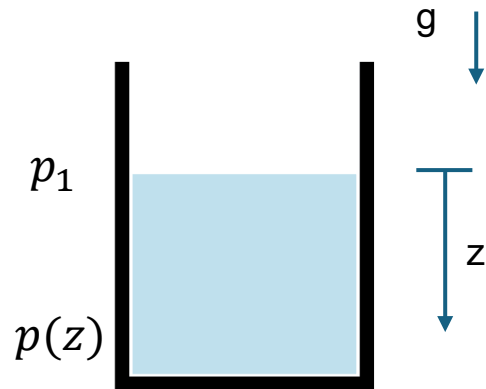
- Thermal equilibrium: two objects at the same temperature.
- 0th law:



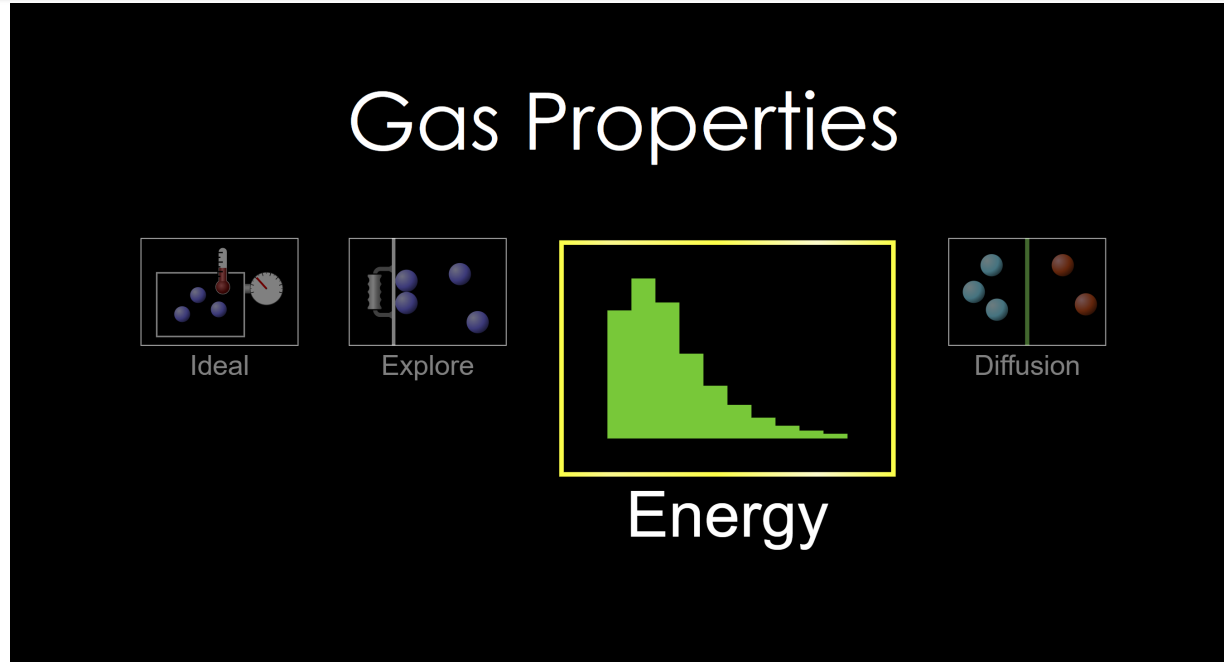
Pressure

- Normal force per unit area $p = \frac{F}{A} = [Pa] = [N/m^2]$
- Absolute pressure p_{abs} relative to vacuum
- Gauge pressure: $p_g = p_{abs} - p_{atm}$
- Vacuum pressure: $p_g = p_{atm} - p_{abs}$
- External forces:

$$p(z) = p_1 + \rho g \Delta z$$

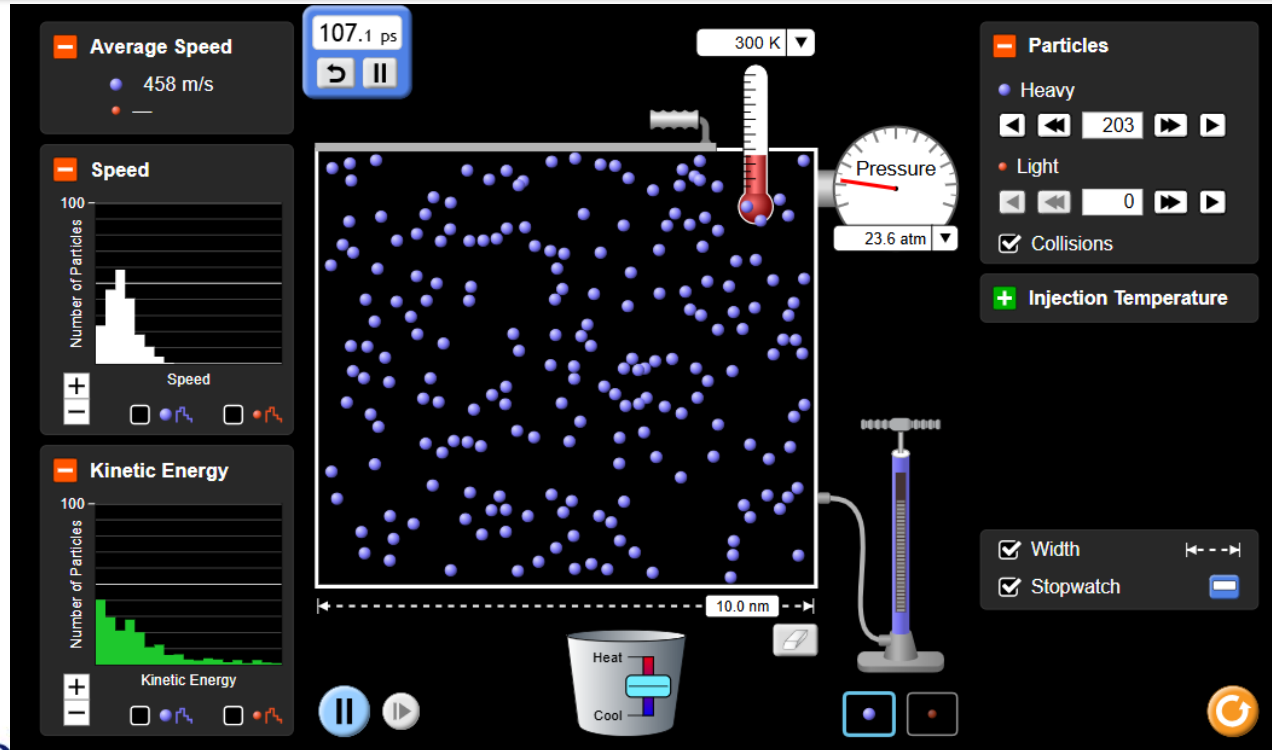


Simulation: Gas properties



https://phet.colorado.edu/sims/html/gas-properties/latest/gas-properties_all.html

Simulation: Gas properties



Group Work: in groups of 2; discuss the following question

Given:

Molecular dynamics simulator for a simple gaseous system

Question:

1. What instruments do you need to define the state of the system?

Simulator



PolLEV



To Do:

- Explore the workings of the Simulator
- Go to pollev.com/claudiohail
- Enter the first names of your group members, e.g. Olivia and Liam.
- ... and answer question 1.

https://phet.colorado.edu/sims/html/gas-properties/latest/gas-properties_all.html

Energy

- An abstract mathematical idea. Energy is always conserved, cannot be generated, destroyed – rather converted from one form to another.
- No information on absolute energy rather changes in energy.
- Can be transferred from one system to another.

Forms of Energy

- Different forms of Energy:

Mechanical
Energy

- Kinetic energy KE

- Potential energy PE

- Internal energy U

- Electromagnetic energy

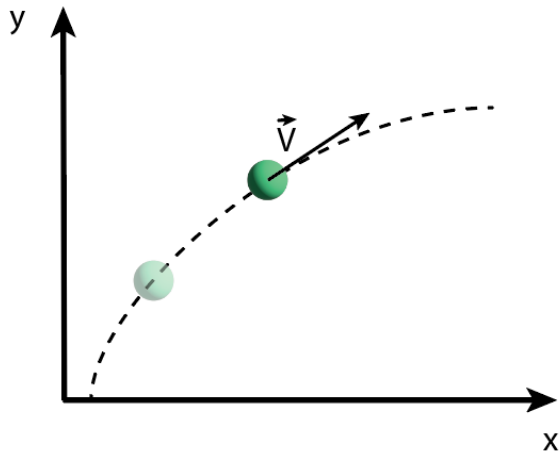
- Binding energy in nucleus of atoms (weak, strong)

$$E = U + KE + PE + \dots \qquad e = \frac{E}{m} = u + ke + pe + \dots$$

$$\Delta E = \Delta U + \Delta KE + \Delta PE + \dots \qquad \dot{E} = \dot{m}e$$

Kinetic energy

- Energy as a result of motion with respect to a reference frame.



Example of linear translation motion of a ball

$$KE = m \frac{|\vec{V}|^2}{2} = m \frac{V^2}{2} = [kJ]$$

Purely mechanical form of energy

Can be:

- Translation
- Rotation
- Vibration

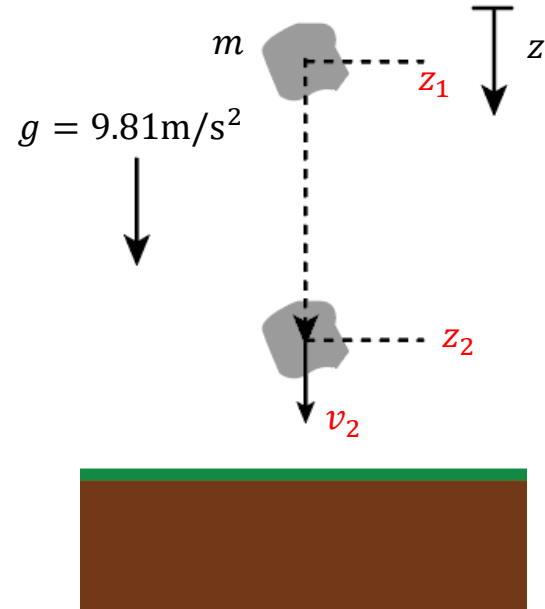
Can be directly converted to work!

Potential energy

- Energy held by an object subject to a conservative force field (potential).
- Gravity:

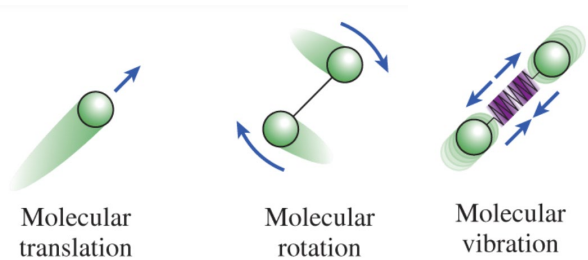
$$PE = mgz$$

$$\Delta PE = mg(z_2 - z_1)$$



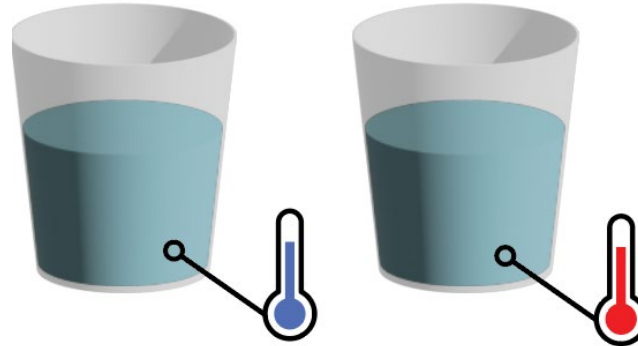
Internal Energy

- Microscopic nature of energy originating in the microscopic kinetic energy of matter (translational, rotational, vibrational), binding energy and chemical energy (potential).

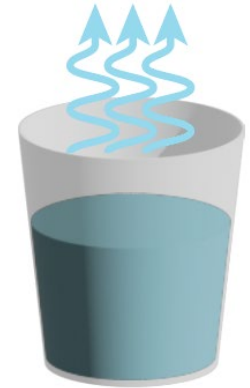


The higher KE the
higher U

UC Berkeley



Sensible energy



Latent energy
Intermolecular
binding forces

Energy transfer by heat

- Way of transferring energy to a closed system.
- Transfer of energy by virtue of a temperature difference. Direction of heat transfer from hot to cold.

- Terminology

- Heat transfer

$$Q \text{ [J]}$$

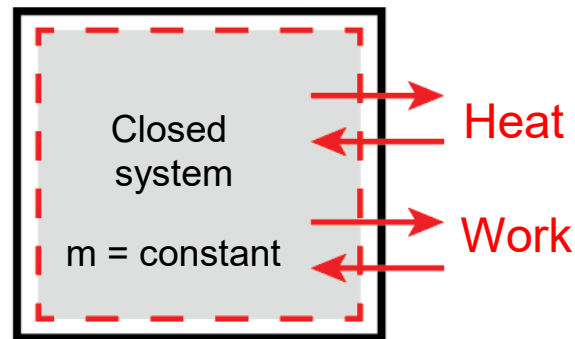
- Heat transfer rate:

$$\dot{Q} \text{ [W]}$$

- Heat transfer per mass:

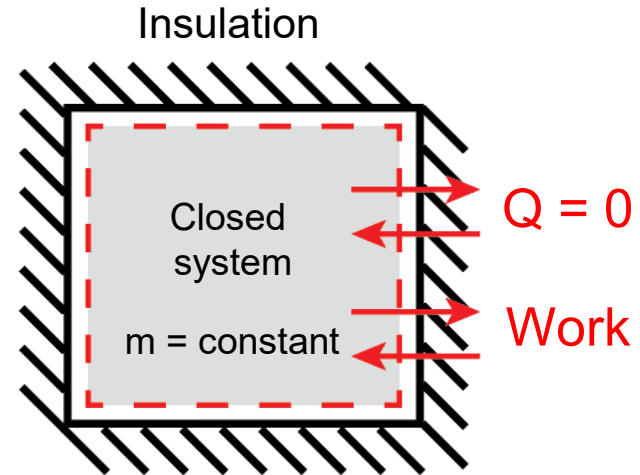
$$q = \frac{Q}{m} \left[\frac{\text{J}}{\text{kg}} \right]$$

$$Q_{12} = \int_{t_1}^{t_2} \dot{Q} dt$$



Adiabatic process

- A process where a system does not exchange heat with the surroundings.
- *Adiabatos*: not to be passed.



Summary

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